

EBN111/122

Tutorial 1 – Memo 21 July 2017

Electric current
is the time rate of
change in charge, measured
in amperes (A)

$$i = \frac{dq}{dt}$$

$$i [A]$$

$$q [C]$$

$$t [s]$$



1.4 A total charge of 300 C flows past a given cross section of a conductor in 30 seconds. What is the value of the current?

$$q = 300 \text{ C} \quad t = 30 \text{ s}$$

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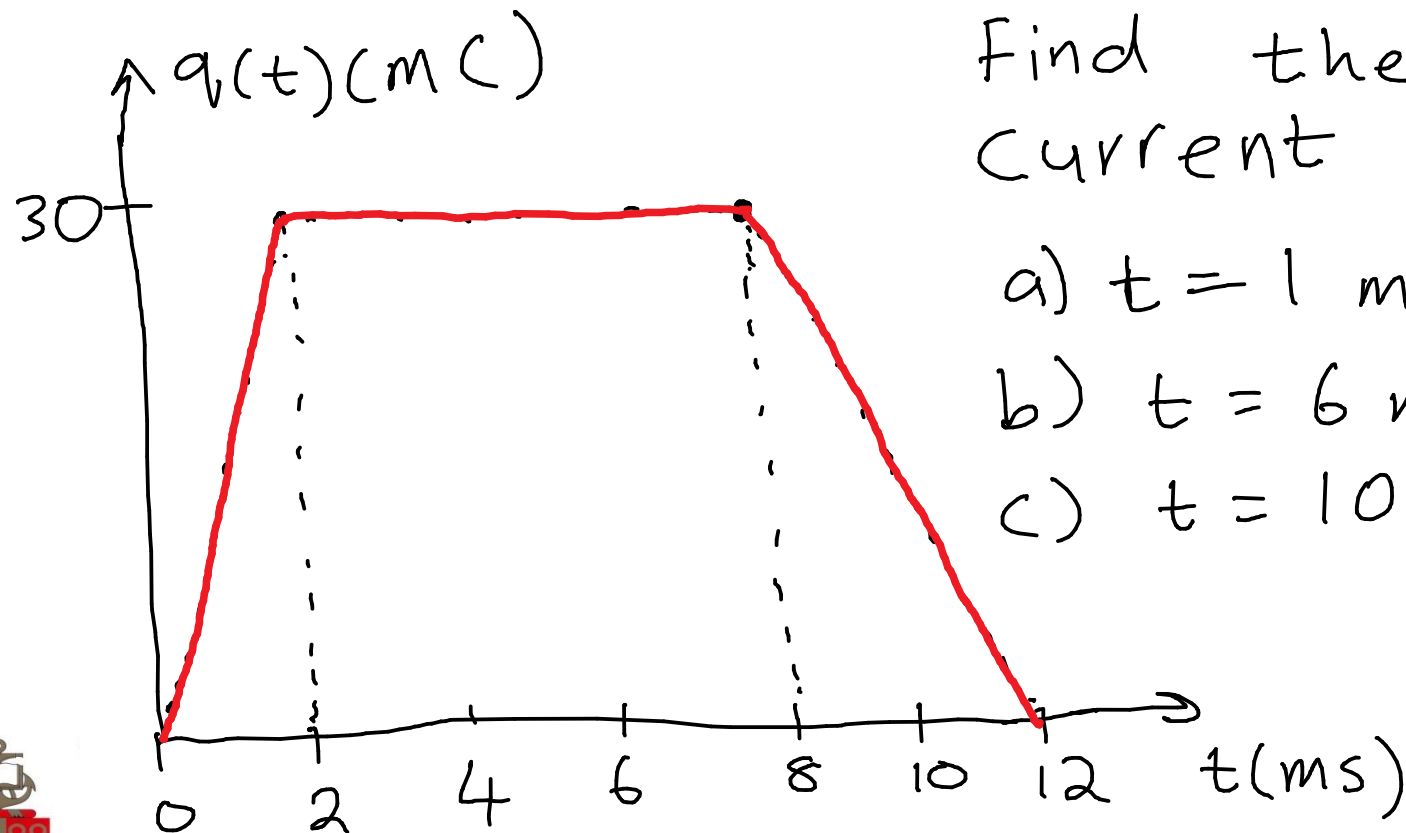
$$i = \frac{dq}{dt}$$

$$= \frac{\Delta q}{\Delta t} \quad * \text{ DC condition}$$

$$= \frac{300 \text{ C}}{30 \text{ s}}$$

$$= 10 \text{ A}$$

1.6 The charge entering a certain element is as follows:



Find the current at:

a) $t = 1$ ms

b) $t = 6$ ms

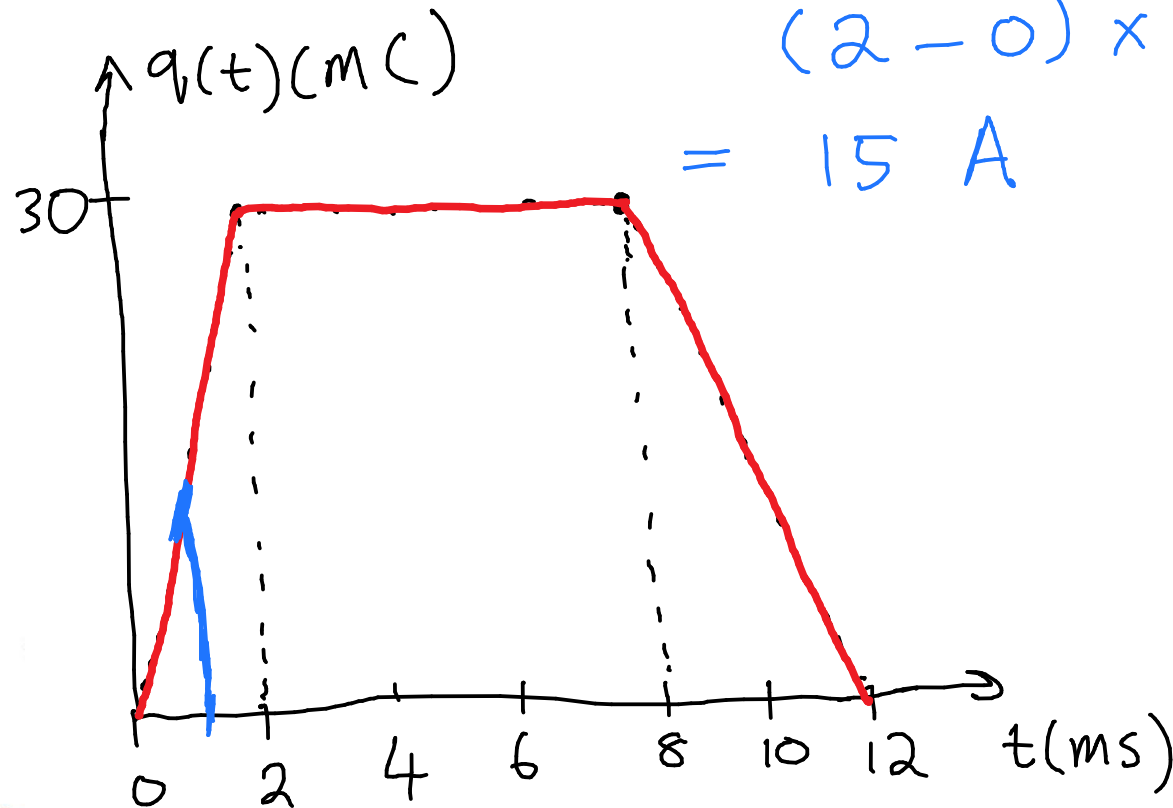
c) $t = 10$ ms

$$a) t = 1 \text{ ms}$$

$$i = \frac{\Delta q}{\Delta t}$$

$$= \frac{(30 - 0) \times 10^{-3} \text{ C}}{(2 - 0) \times 10^{-3} \text{ s}}$$

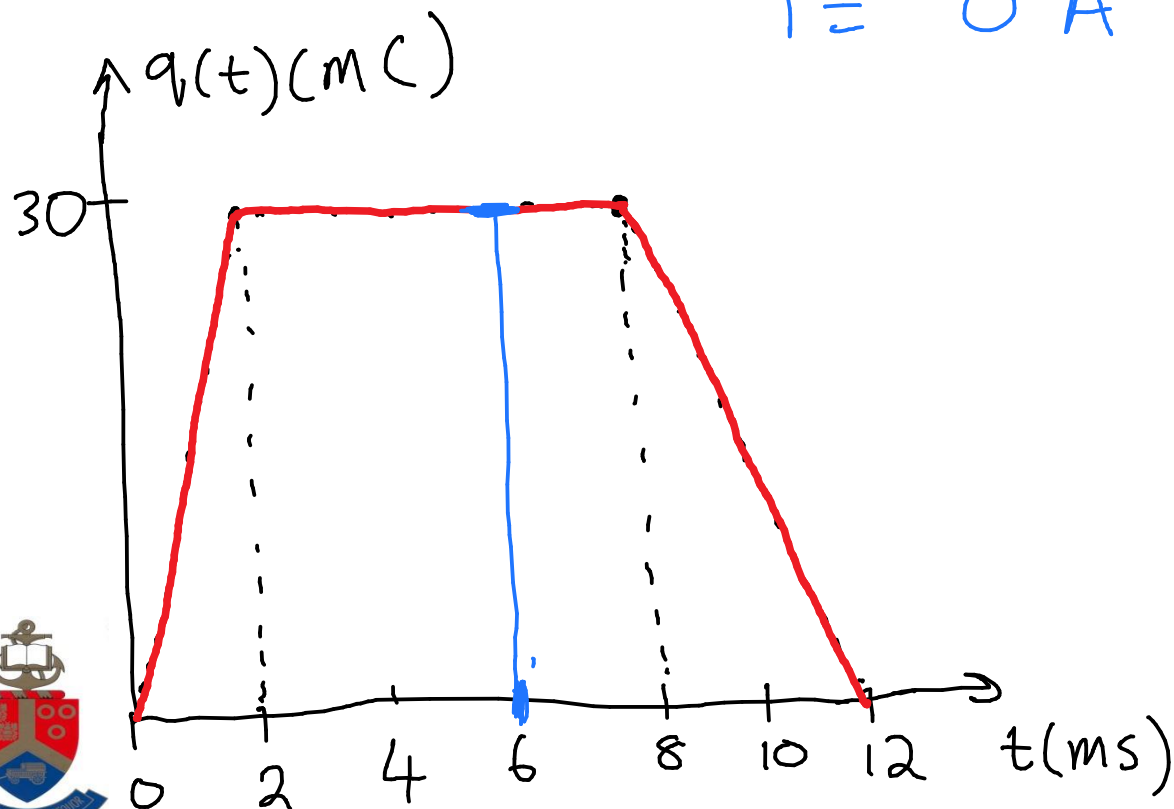
$$= 15 \text{ A}$$



b) $t = 6 \text{ ms}$

no change in charge,
therefore no current

$i = 0 \text{ A}$

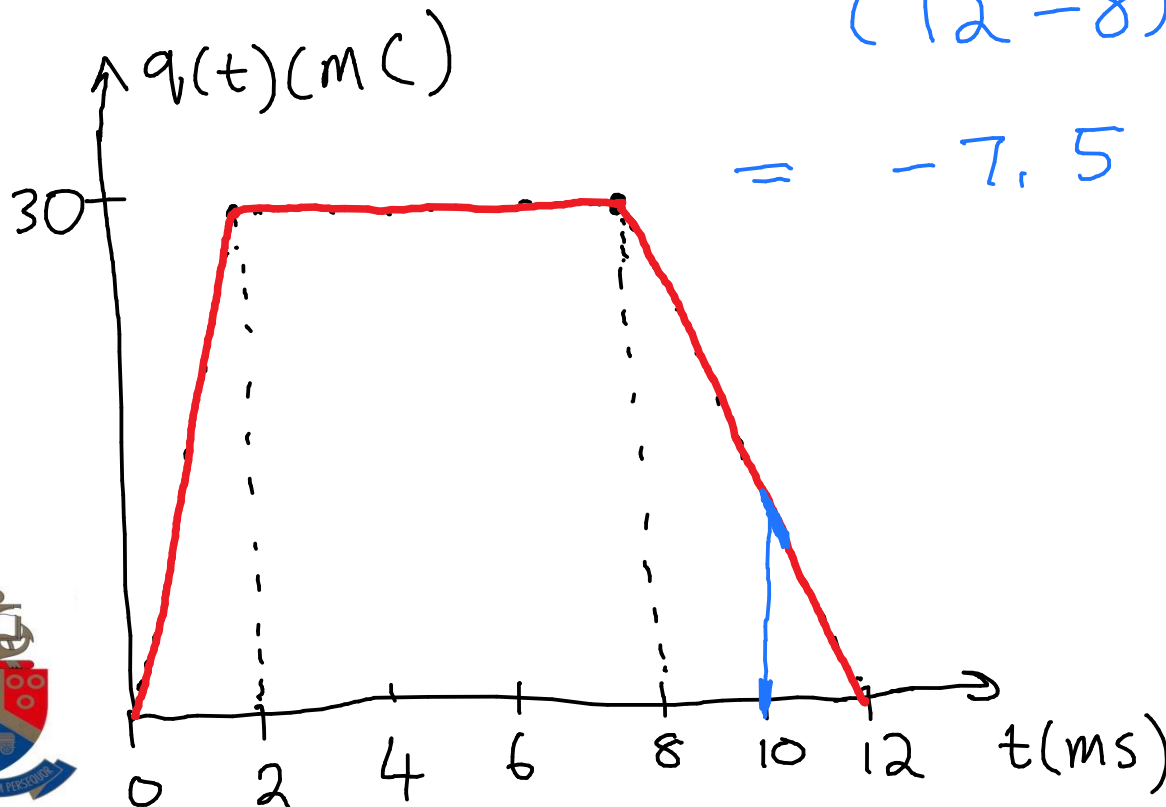


c) $t = 10 \text{ ms}$

$$i = \frac{\Delta q}{\Delta t}$$

$$= \frac{(0 - 30) \times 10^{-3} \text{ C}}{(12 - 8) \times 10^{-3} \text{ s}}$$

$$= -7.5 \text{ A}$$



The equation relating current to charge is given by:

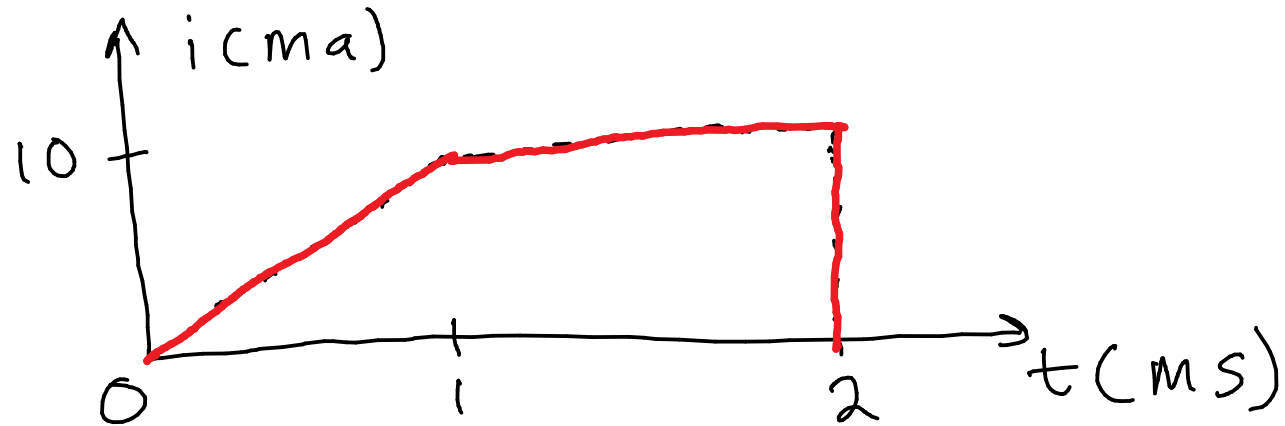
$$i = \frac{dq}{dt},$$

integrating both sides, allows us to find the charge transferred between time t_0 and t .

$$Q = \int_{t_0}^t i d\tau$$



1.8 The current flowing past a point in a device is shown.



Calculate the total charge through the point.



Q = area under current graph

$$= \frac{1}{2} b \cdot h + b \cdot h$$

$$= 0.5 \times 1 \times 10^{-3} \times 10 \times 10^{-3} +$$

$$1 \times 10^{-3} \times 10 \times 10^{-3}$$

$$= 15 \mu C$$

Power is the time rate
of expanding or absorbing
energy, measured in watts (W).

$$P = \frac{dw}{dt}$$

$$P = V \times i$$

$$W = \int_{t_0}^t P dt$$

$$P [W]$$

$$W [J] [Wh]$$

$$\neq 1 Wh = 3600 J$$

$$t [s]$$

$$V [V]$$

$$i [A]$$



1.11 A rechargeable

flashlight is capable of
delivering 90 mA for 12 hours.

a) How much charge can it
release at that rate?

b) If its terminal voltage is
1.5 V, how much energy can
it deliver?

$$i = 90 \times 10^{-3} \text{ A}$$

$$v = 1.5 \text{ V}$$

$$t = 12 \text{ h} \\ = 43200 \text{ s}$$



$$i = 90 \times 10^{-3} \text{ A} \quad t = 43200 \text{ s}$$

$$q = \int_{t_0}^t i dt$$

$$= i \times t \quad * \text{ DC conditions}$$

$$= (90 \times 10^{-3} \text{ A})(43200 \text{ s})$$

$$= 3888 \text{ C}$$

$$i = 90 \times 10^{-3} \text{ A}$$

$$V = 1.5 \text{ V}$$

$$t = 43200 \text{ s}$$

$$W = \int_{t_0}^t p d\tau = \int_{t_0}^t v i d\tau$$

$$= V \times i \times t \quad \text{* DC conditions}$$

$$= (1.5 \text{ V}) \times (90 \times 10^{-3} \text{ A}) \times (43200 \text{ s})$$

$$= 5832 \text{ J}$$



$$i = 90 \times 10^{-3} \text{ A}$$

$$V = 1.5 \text{ V}$$

$$t = 12 \text{ h}$$

$$W = \int_{t_0}^t p d\tau = \int_{t_0}^t v i d\tau$$

$$= V \times i \times t \quad \text{* DC conditions}$$

$$= (1.5 \text{ V}) \times (90 \times 10^{-3} \text{ A}) \times (12 \text{ h})$$

$$= 1.62 \text{ Wh}$$

$$= 1.62 \text{ Wh} \times 3600 \text{ J/Wh}$$

$$= 5832 \text{ J}$$



1.12 If the current flowing through an element is given by:

$$i(t) \begin{cases} 3t \text{ A,} & 0 \leq t < 6 \text{ s} \\ 18 \text{ A,} & 6 \leq t < 10 \text{ s} \\ -12 \text{ A,} & 10 \leq t < 15 \text{ s} \\ 0 \text{ A,} & 15 \leq t \end{cases}$$

Plot the charge stored in the element over $0 < t < 20 \text{ s}$.

For $0 \leq t < 6$ s

$$q = \int_0^t i d\tau + q(0)$$

$$= \int_0^t 3\tau d\tau + 0$$

$$= \left. \frac{3\tau^2}{2} \right|_0^t$$

$$= \frac{3}{2} t^2 \text{ C}$$

$$q(6) = \frac{3}{2} (6^2) = 54 \text{ C}$$



For $6 \leq t < 10$ s

$$q = \int_6^t i d\tau + q(6)$$

$$= \int_6^t 18 d\tau + 54$$

$$= 18\tau \Big|_6^t + 54$$

$$= 18t - 18(6) + 54$$

$$= 18t - 54 \text{ C}$$

$$q(10) = 18(10) - 54 = 126 \text{ C}$$



For $10 \leq t < 15$ s

$$q = \int_{10}^t i d\tau + q(10)$$

$$= \int_{10}^t -12 d\tau + q(10)$$

$$= -12\tau \Big|_{10}^t + 126$$

$$= -12t - (-12 \times 10) + 126$$

$$= -12t + 246$$

$$q(15) = -12(15) + 246 = 66 \text{ C}$$



For $15 \leq t < 20$ s

$$q = \int_{15}^t i d\tau + q(15)$$

$$= \int_{15}^t 0 d\tau + 66$$

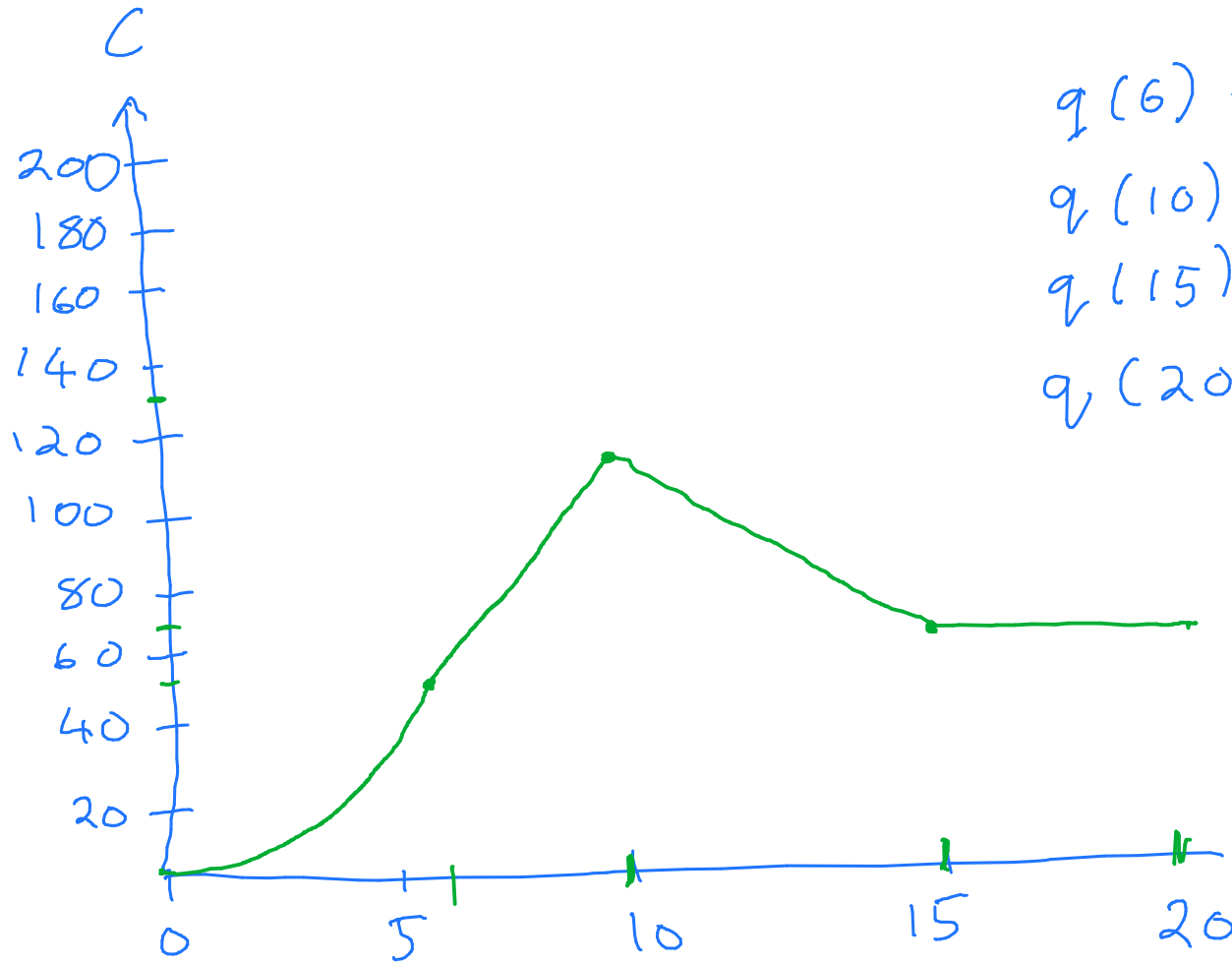
$$= 66 \text{ C}$$

$$q(20) = 66 \text{ C}$$



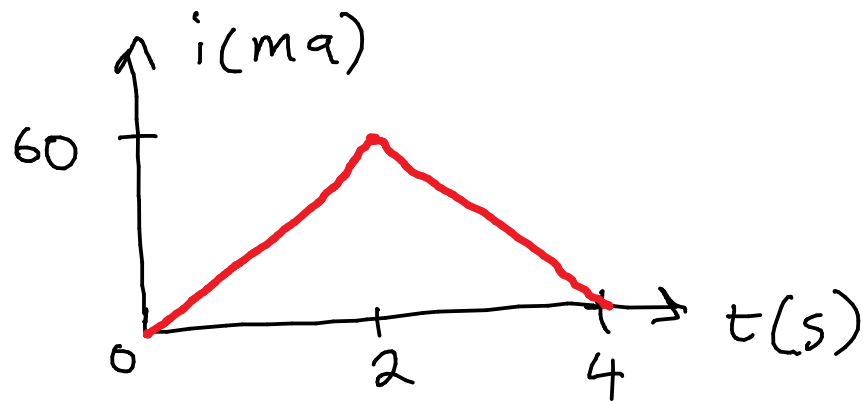
$\therefore$ For $0 \leq t < 20s$

$$q(t) \begin{cases} 1,5t^2 \text{ C} & 0 \leq t < 6s \\ 18t - 54 \text{ C} & 6 \leq t < 10s \\ -12t + 246 \text{ C} & 10 \leq t < 15s \\ 66 \text{ C} & 15s \leq t \end{cases}$$



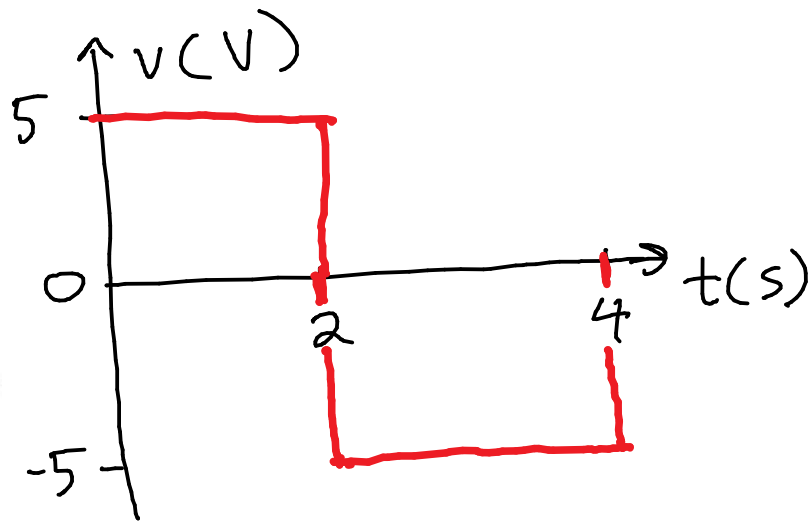
$$\begin{aligned}q(6) &= 54\text{ C} \\q(10) &= 126\text{ C} \\q(15) &= 66\text{ C} \\q(20) &= 66\text{ C}\end{aligned}$$

1.16 The current through and the voltage across an element are shown.

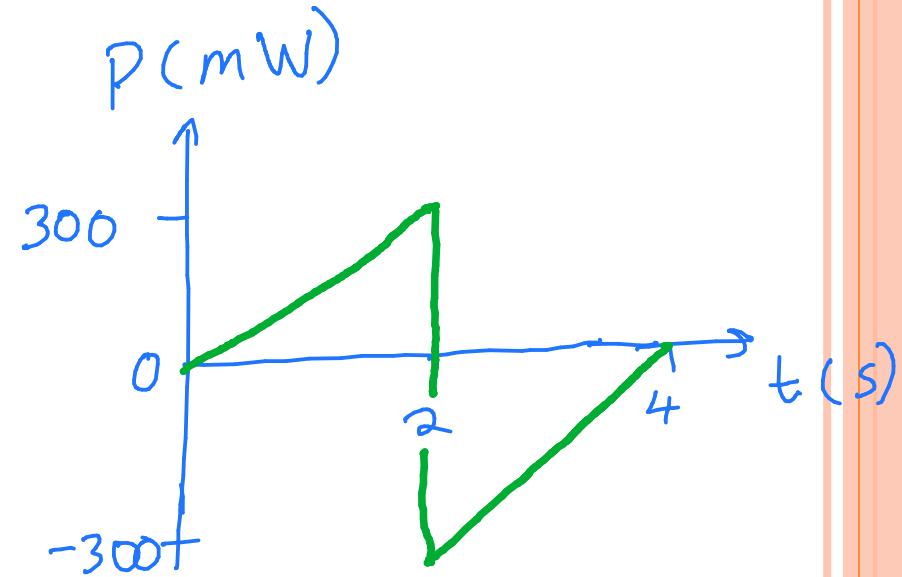
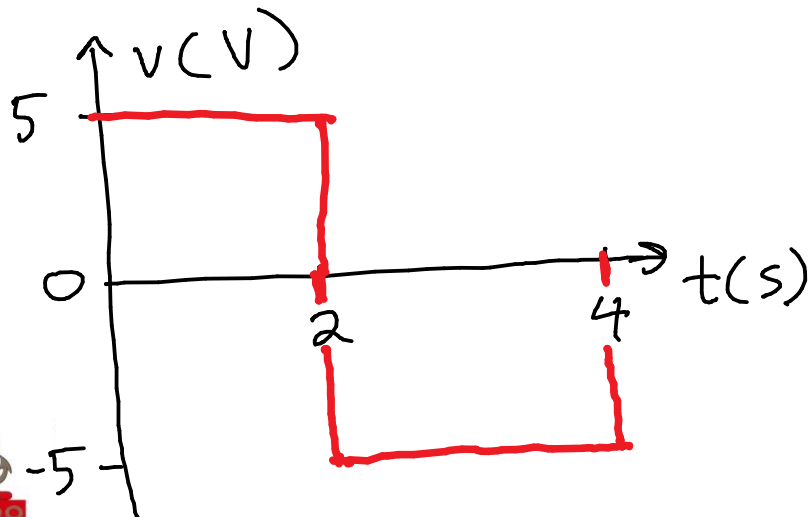
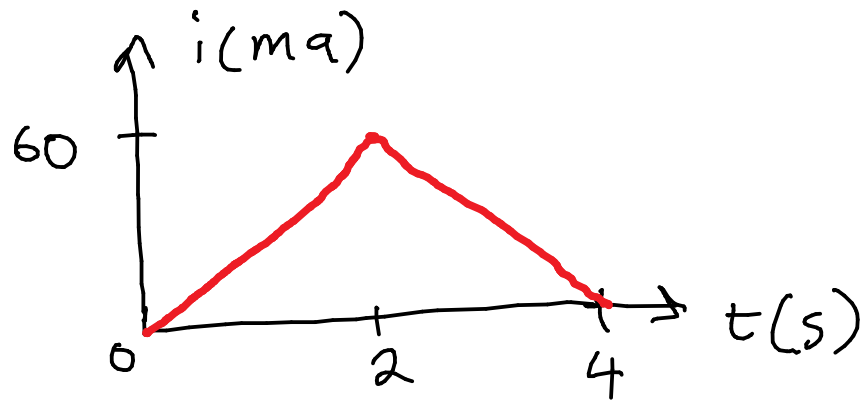


a) Sketch the power delivered for $t > 0$.

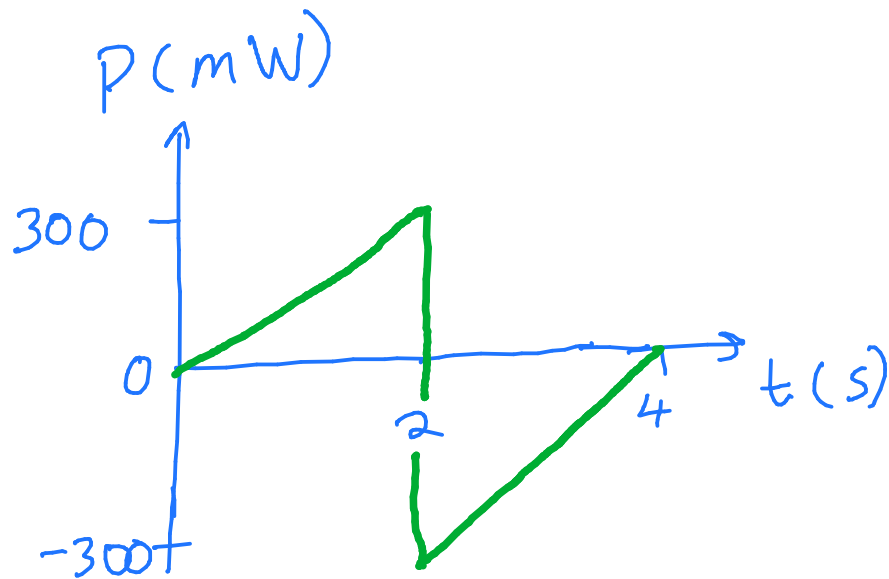
b) Find the total energy absorbed for $0 < t < 4$.



a) Power delivered



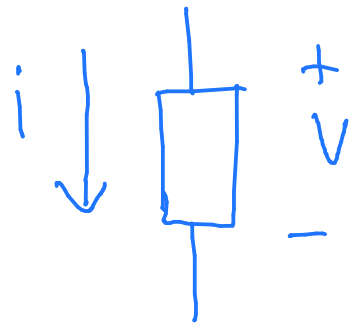
b) Energy absorbed



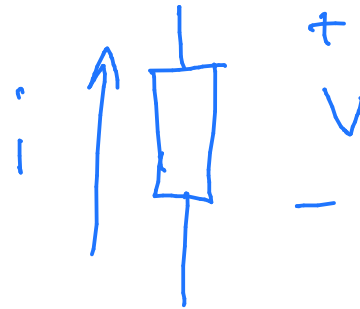
$$\begin{aligned} w &= \text{net area} \\ &= \frac{1}{2} b \cdot h - \frac{1}{2} b \cdot h \\ &= \frac{1}{2} \times 2 \times 0.3 - \\ &\quad \frac{1}{2} \times 2 \times 0.3 \\ &= 0 \text{ W} \end{aligned}$$

Passive sign convention

power is absorbed when current enters through positive terminal, $p = +Vi$, and negative when current enters through negative terminal, $p = -Vi$.



$$p = +Vi$$



$$p = -Vi$$



law of conservation of energy.

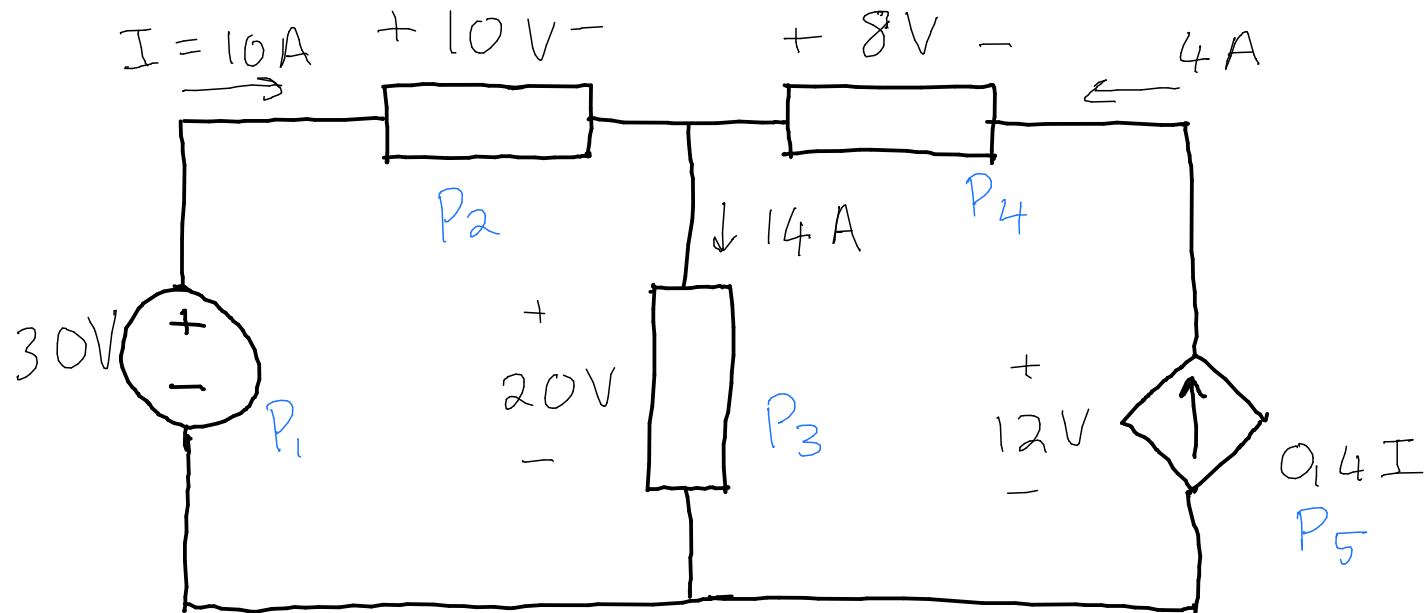
The algebraic sum of power in a circuit, is always zero.

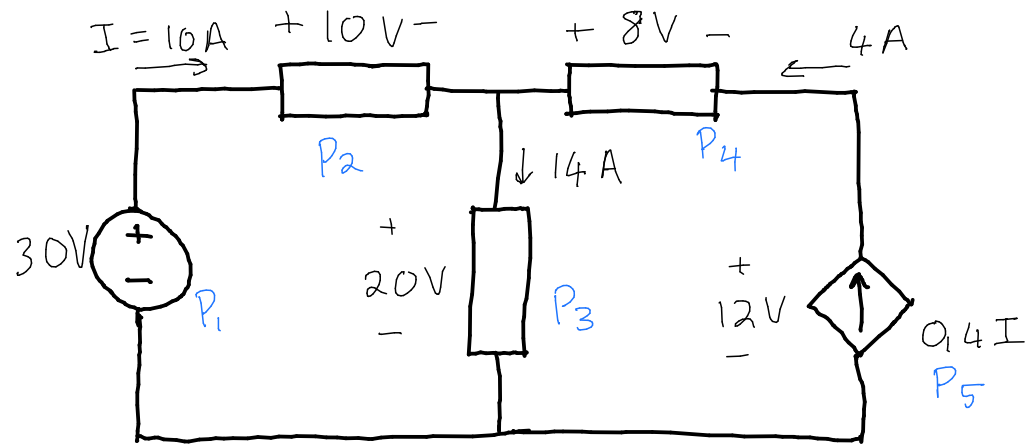
$$\sum p = 0$$

$$p [W]$$



1.18 Find the power absorbed by each element.





$$P_1 = 30(-10) = -300 \text{ W}$$

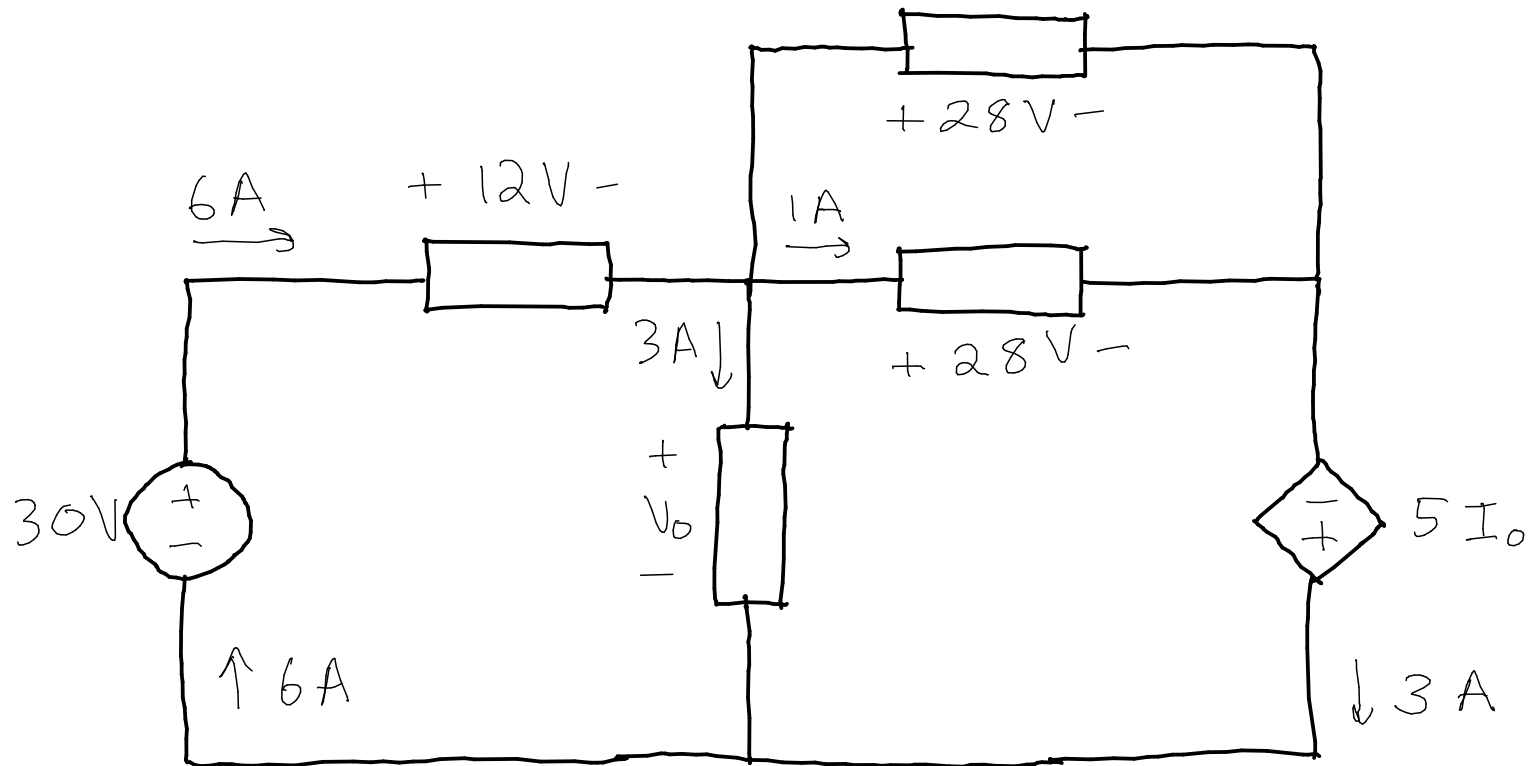
$$P_2 = 10(10) = 100 \text{ W}$$

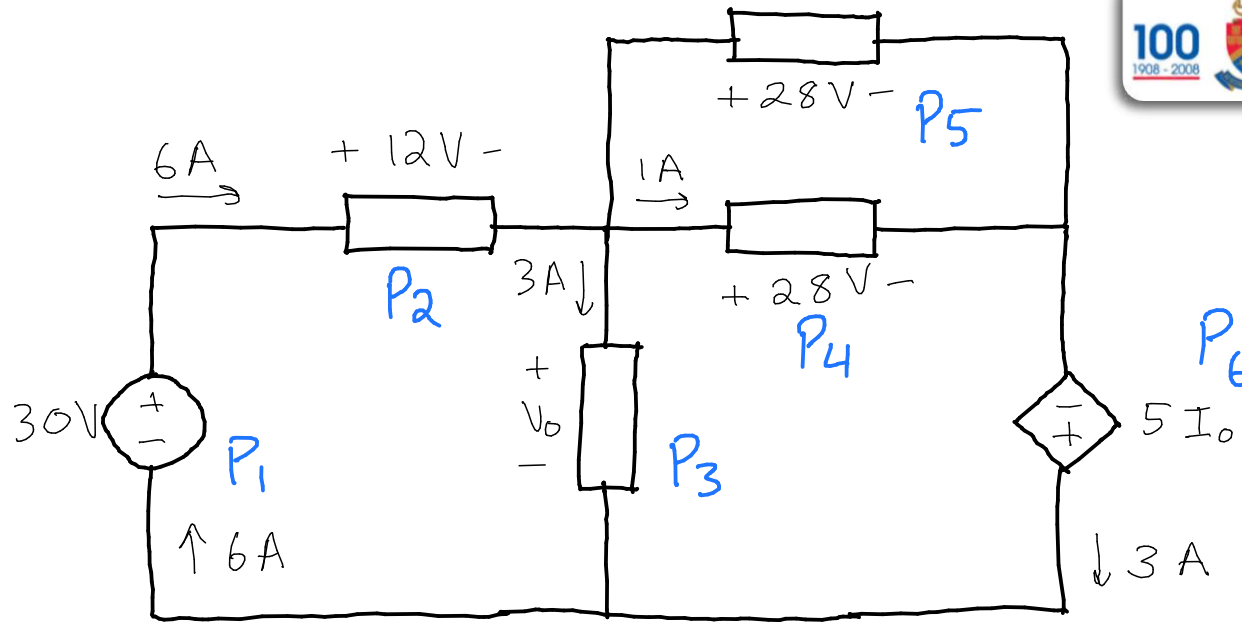
$$P_3 = 20(14) = 280 \text{ W}$$

$$P_4 = 8(-4) = -32 \text{ W}$$

$$P_5 = 12(-0.4 \times 10) = -48 \text{ W}$$

1.20 Find V_o and the power absorbed by each element in the circuit.





$$P_1 = 30(-6) = -180 \text{ W}$$

$$P_4 = 28(1) = 28 \text{ W}$$

$$P_2 = 12(6) = 72 \text{ W}$$

$$P_5 = 28(2) = 56 \text{ W}$$

$$P_3 = V_0(3)$$

$$P_6 = 5(2)(-3) = -30 \text{ W}$$

$$P_1 = -180 \text{ W}$$

$$P_2 = 72 \text{ W}$$

$$P_3 = V_0(3)$$

$$P_4 = 28 \text{ W}$$

$$P_5 = 56 \text{ W}$$

$$P_6 = -30 \text{ W}$$

$$\sum P = 0$$

$$-180 + 72 + P_3 + 28 + 56 - 30 = 0$$

$$P_3 = 54 \text{ W}$$



$$P_3 = 54 \text{ W}$$

$$P_3 = V_o(3)$$

$$54 = V_o(3)$$

$$V_o = \frac{54}{3}$$

$$V_o = 18 \text{ V}$$



1.24 A utility company charges 8.2 cents / kWh.

If a consumer operates a 60 W light bulb continuously for one day, how much is the consumer charged?

$$\text{price} = 8.2 \text{ cents / kWh}$$

$$P = 60 \text{ W}$$

$$t = 24 \text{ h}$$



price = 8.2 cents/kWh

$P = 60 \text{ W}$

$t = 24 \text{ h}$

$$W = \int_{t_0}^t P d\tau$$

$$= P \times t \quad * \text{DC conditions}$$

$$= 60 \times 24$$

$$= 1440 \text{ Wh}$$

$$\text{bill} = 1.44 \text{ kWh} \times 8.2 \text{ cents/kWh}$$

$$= 11.808 \text{ cents}$$



1.31 In a household, a business is run for an average of 6 h/day. The total power consumed by the computer and printer is 230 W. In addition a 75 W light runs during the same 6 h. If their utility charges 11,75 cents/kWh, how much do the owners pay every 30 days?

$$t = 6 \times 30 \text{ h}$$

$$P = 230 + 75 \text{ W}$$

$$\text{price} = 11,75 \text{ cents/kWh}$$



$$t = 180 \text{ h}$$

$$p = 305 \text{ W}$$

$$\text{price} = 11.75 \text{ cents/KWh}$$

$$w = \int_{t_0}^t p d\tau$$

$$= p \times t \quad * \text{ DC conditions}$$

$$= 180 \text{ h} \times 305 \text{ W}$$

$$= 54.9 \text{ kWh}$$

$$\text{bill} = 54.9 \text{ kWh} \times 11.75 \text{ cents/kWh}$$

$$= 645.075 \text{ cents}$$

